

CodeAlpha_EDA_Project

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use("ggplot")

df = pd.read_csv("/content/WA_Fn-UseC_-HR-Employee-Attrition.csv")
df.head()

{"type": "dataframe", "variable_name": "df"}

df.tail()

{"type": "dataframe"}

df.head(5)

{"type": "dataframe", "variable_name": "df"}

df.shape

(1470, 35)

df.columns

Index(['Age', 'Attrition', 'BusinessTravel', 'DailyRate',
      'Department',
      'DistanceFromHome', 'Education', 'EducationField',
      'EmployeeCount',
      'EmployeeNumber', 'EnvironmentSatisfaction', 'Gender',
      'HourlyRate',
      'JobInvolvement', 'JobLevel', 'JobRole', 'JobSatisfaction',
      'MaritalStatus', 'MonthlyIncome', 'MonthlyRate',
      'NumCompaniesWorked',
      'Over18', 'OverTime', 'PercentSalaryHike', 'PerformanceRating',
      'RelationshipSatisfaction', 'StandardHours',
      'StockOptionLevel',
      'TotalWorkingYears', 'TrainingTimesLastYear',
      'WorkLifeBalance',
      'YearsAtCompany', 'YearsInCurrentRole',
      'YearsSinceLastPromotion',
      'YearsWithCurrManager'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 1470 entries, 0 to 1469
```

```
Data columns (total 35 columns):
```

#	Column	Non-Null Count	Dtype
0	Age	1470 non-null	int64
1	Attrition	1470 non-null	object
2	BusinessTravel	1470 non-null	object
3	DailyRate	1470 non-null	int64
4	Department	1470 non-null	object
5	DistanceFromHome	1470 non-null	int64
6	Education	1470 non-null	int64
7	EducationField	1470 non-null	object
8	EmployeeCount	1470 non-null	int64
9	EmployeeNumber	1470 non-null	int64
10	EnvironmentSatisfaction	1470 non-null	int64
11	Gender	1470 non-null	object
12	HourlyRate	1470 non-null	int64
13	JobInvolvement	1470 non-null	int64
14	JobLevel	1470 non-null	int64
15	JobRole	1470 non-null	object
16	JobSatisfaction	1470 non-null	int64
17	MaritalStatus	1470 non-null	object
18	MonthlyIncome	1470 non-null	int64
19	MonthlyRate	1470 non-null	int64
20	NumCompaniesWorked	1470 non-null	int64
21	Over18	1470 non-null	object
22	Overtime	1470 non-null	object
23	PercentSalaryHike	1470 non-null	int64
24	PerformanceRating	1470 non-null	int64
25	RelationshipSatisfaction	1470 non-null	int64
26	StandardHours	1470 non-null	int64
27	StockOptionLevel	1470 non-null	int64
28	TotalWorkingYears	1470 non-null	int64
29	TrainingTimesLastYear	1470 non-null	int64
30	WorkLifeBalance	1470 non-null	int64
31	YearsAtCompany	1470 non-null	int64
32	YearsInCurrentRole	1470 non-null	int64
33	YearsSinceLastPromotion	1470 non-null	int64
34	YearsWithCurrManager	1470 non-null	int64

```
dtypes: int64(26), object(9)
```

```
memory usage: 402.1+ KB
```

```
df.describe()
```

```
{"type": "dataframe"}
```

```
df.describe(include="object")
```

```
{
  "summary": {
    "name": "df",
    "rows": 4,
    "fields": [
      {
        "column": "Attrition",
        "properties": {
          "dtype": "string",
          "num_unique_values": 4,
          "samples": [
            2,
            "1233",
            "1470"
          ],
          "semantic_type": "",
          "description": ""
        },
        "column": "BusinessTravel",
        "properties": {
          "dtype": "string",
          "num_unique_values": 4,
          "samples": [
            3,
            "1043",
            "1470"
          ],
          "semantic_type": "",
          "description": ""
        },
        "column": "Department",
        "properties": {
          "dtype": "string",
          "num_unique_values": 4,
          "samples": [
            3,
            "961",
            "1470"
          ],
          "semantic_type": "",
          "description": ""
        },
        "column": "EducationField",
        "properties": {
          "dtype": "string",
          "num_unique_values": 4,
          "samples": [
            6,
            "606",
            "1470"
          ],
          "semantic_type": "",
          "description": ""
        },
        "column": "Gender",
        "properties": {
          "dtype": "string",
          "num_unique_values": 4,
          "samples": [
            2,
            "882",
            "1470"
          ],
          "semantic_type": "",
          "description": ""
        },
        "column": "JobRole",
        "properties": {
          "dtype": "string",
          "num_unique_values": 4,
          "samples": [
            9,
            "326",
            "1470"
          ],
          "semantic_type": "",
          "description": ""
        },
        "column": "MaritalStatus",
        "properties": {
          "dtype": "string",
          "num_unique_values": 4,
          "samples": [
            3,
            "673",
            "1470"
          ],
          "semantic_type": "",
          "description": ""
        },
        "column": "Over18",
        "properties": {
          "dtype": "string",
          "num_unique_values": 3,
          "samples": [
            "1470",
            1,
            "Y"
          ],
          "semantic_type": "",
          "description": ""
        },
        "column": "OverTime",
        "properties": {
          "dtype": "string",
          "num_unique_values": 4,
          "samples": [
            2,
            "1054",
            "1470"
          ],
          "semantic_type": "",
          "description": ""
        }
      ]
    }
  },
  "type": "dataframe"
}
```

```
df.isnull().sum()
```

Age	0
Attrition	0
BusinessTravel	0
DailyRate	0
Department	0

```

DistanceFromHome      0
Education              0
EducationField         0
EmployeeCount          0
EmployeeNumber         0
EnvironmentSatisfaction 0
Gender                 0
HourlyRate             0
JobInvolvement         0
JobLevel               0
JobRole                0
JobSatisfaction        0
MaritalStatus          0
MonthlyIncome          0
MonthlyRate            0
NumCompaniesWorked     0
Over18                 0
OverTime               0
PercentSalaryHike      0
PerformanceRating      0
RelationshipSatisfaction 0
StandardHours          0
StockOptionLevel       0
TotalWorkingYears      0
TrainingTimesLastYear  0
WorkLifeBalance        0
YearsAtCompany         0
YearsInCurrentRole     0
YearsSinceLastPromotion 0
YearsWithCurrManager   0
dtype: int64

df.duplicated().sum()

np.int64(0)

df = df.drop_duplicates()

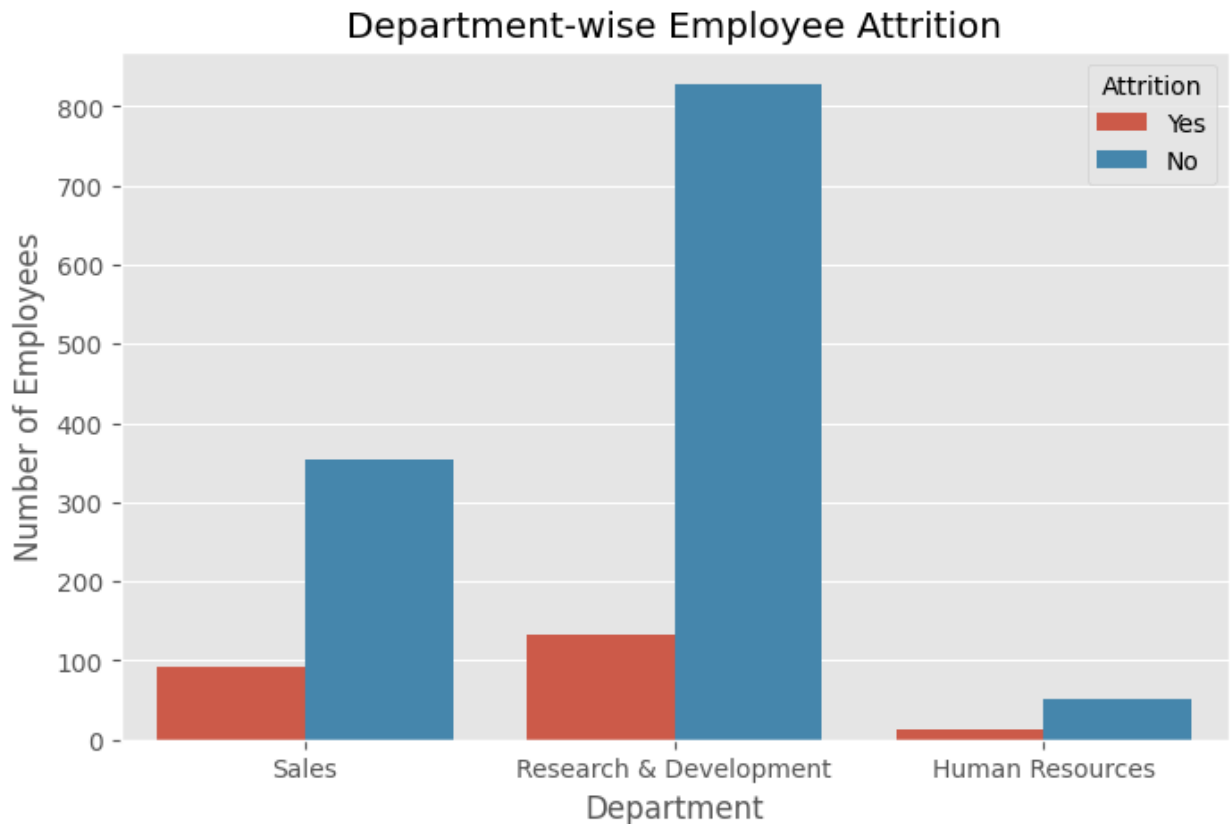
```

1. Which department has the highest employee attrition?

```

plt.figure(figsize=(8,5))
sns.countplot(data=df, x='Department', hue='Attrition')
plt.title('Department-wise Employee Attrition')
plt.xlabel('Department')
plt.ylabel('Number of Employees')
plt.show()

```



Research & Development (R&D) has the highest number of employees leaving the organization, followed by Sales. This suggests that HR should investigate workload, employee engagement, and career growth opportunities within the R&D department to reduce attrition.

```
# Calculate attrition rate by department
```

```
department_attrition = (  
    df.groupby('Department')['Attrition']  
      .apply(lambda x: (x == 'Yes').mean() * 100)  
      .sort_values(ascending=False)  
)
```

```
print(department_attrition)
```

```
Department  
Sales                20.627803  
Human Resources      19.047619  
Research & Development 13.839750  
Name: Attrition, dtype: float64
```

```
plt.figure(figsize=(8,5))
```

```
sns.barplot(  

```

```

x=department_attrition.index,
y=department_attrition.values,
palette='Set2'
)

plt.title('Department-wise Attrition Rate (%)')
plt.xlabel('Department')
plt.ylabel('Attrition Rate (%)')

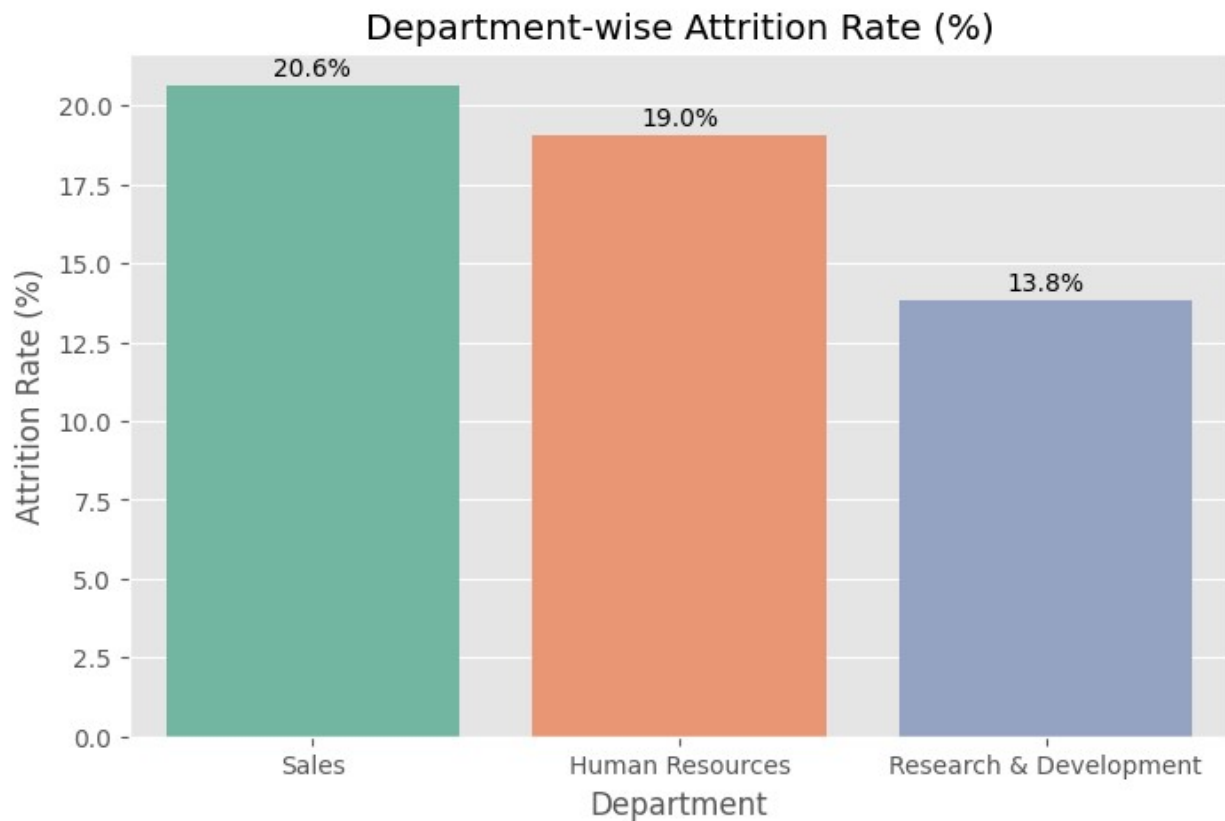
# Add percentage labels
for i, value in enumerate(department_attrition.values):
    plt.text(i, value + 0.3, f'{value:.1f}%', ha='center')

plt.show()

/tmp/ipykernel_7919/689596041.py:3: FutureWarning:
Passing `palette` without assigning `hue` is deprecated and will be
removed in v0.14.0. Assign the `x` variable to `hue` and set
`legend=False` for the same effect.

sns.barplot(

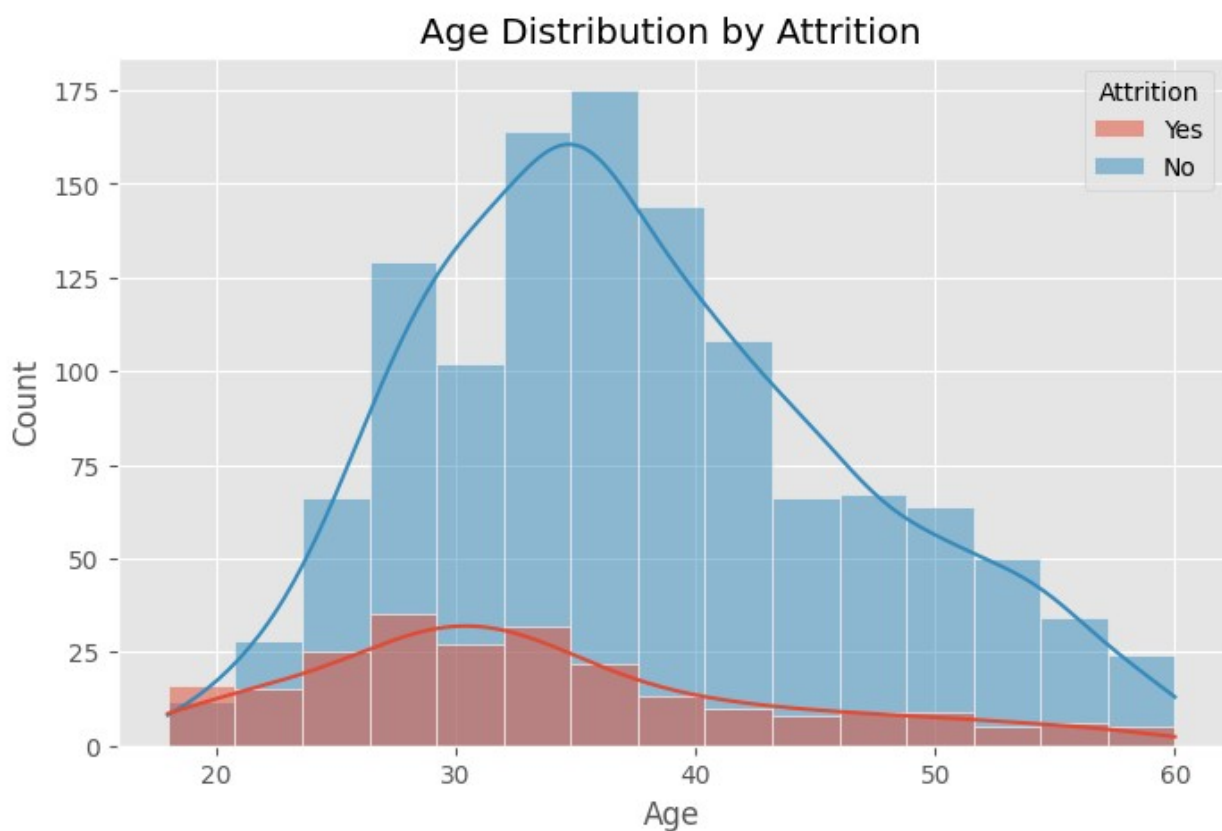
```



Although the Research & Development (R&D) department has the highest number of employees leaving, the Sales department has the highest attrition rate (%). This means employees in the Sales department are more likely to leave relative to the size of that department. Therefore, retention initiatives should prioritize the Sales department while also addressing the larger absolute number of exits in R&D.

Which age group has the highest employee attrition?

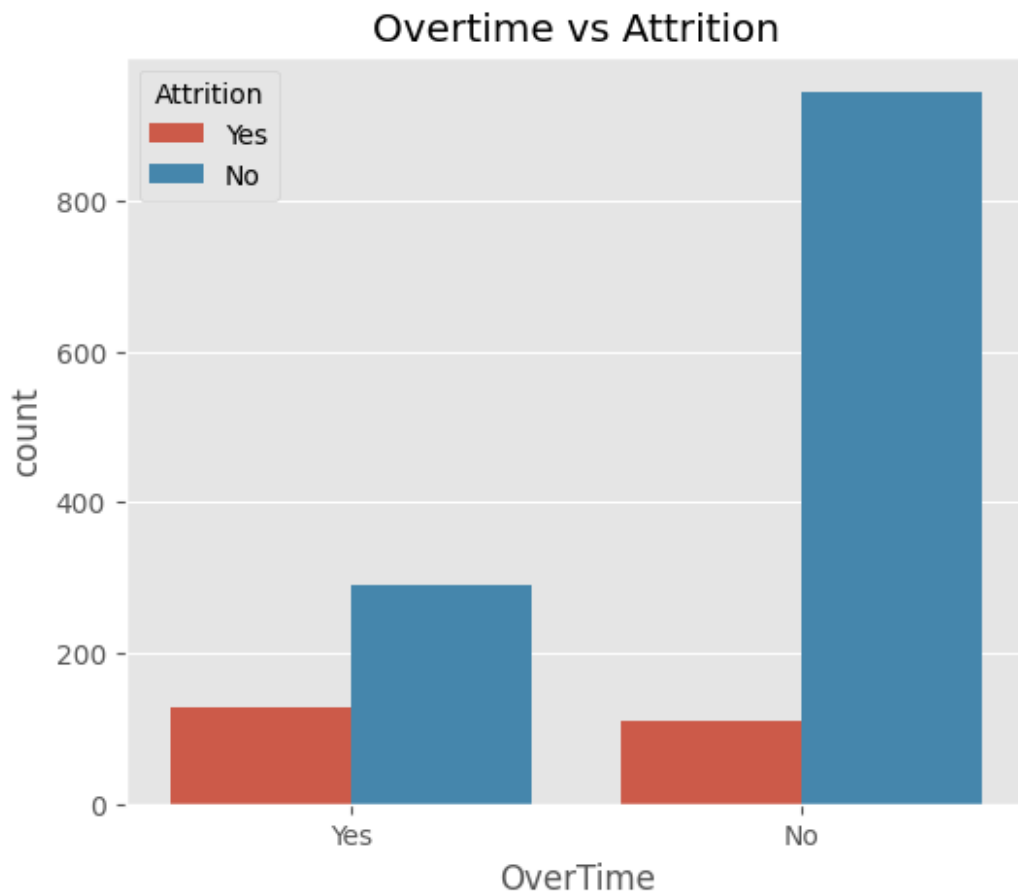
```
plt.figure(figsize=(8,5))
sns.histplot(data=df, x='Age', hue='Attrition', bins=15, kde=True)
plt.title('Age Distribution by Attrition')
plt.show()
```



Employees between 25–35 years show the highest attrition, indicating younger professionals are more likely to change jobs for career growth

Does overtime increase employee attrition?

```
plt.figure(figsize=(6,5))
sns.countplot(data=df, x='OverTime', hue='Attrition')
plt.title('Overtime vs Attrition')
plt.show()
```



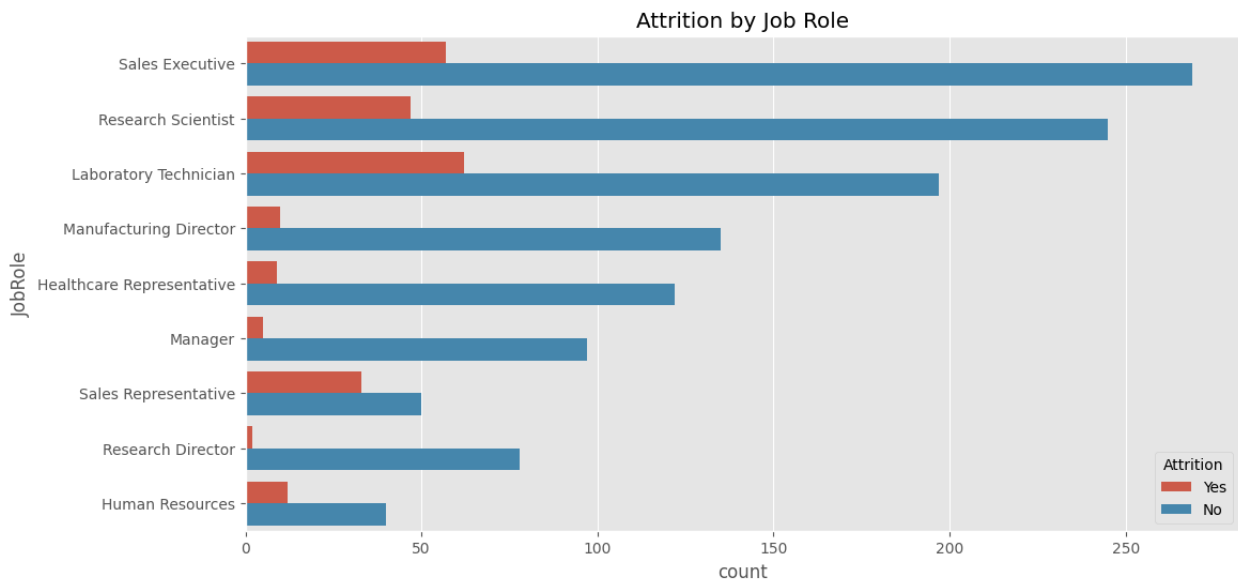
Employees who frequently work overtime have a significantly higher attrition rate than those who do not.

Which job role has the highest attrition?

```
plt.figure(figsize=(12,6))
sns.countplot(data=df, y='JobRole', hue='Attrition')
```



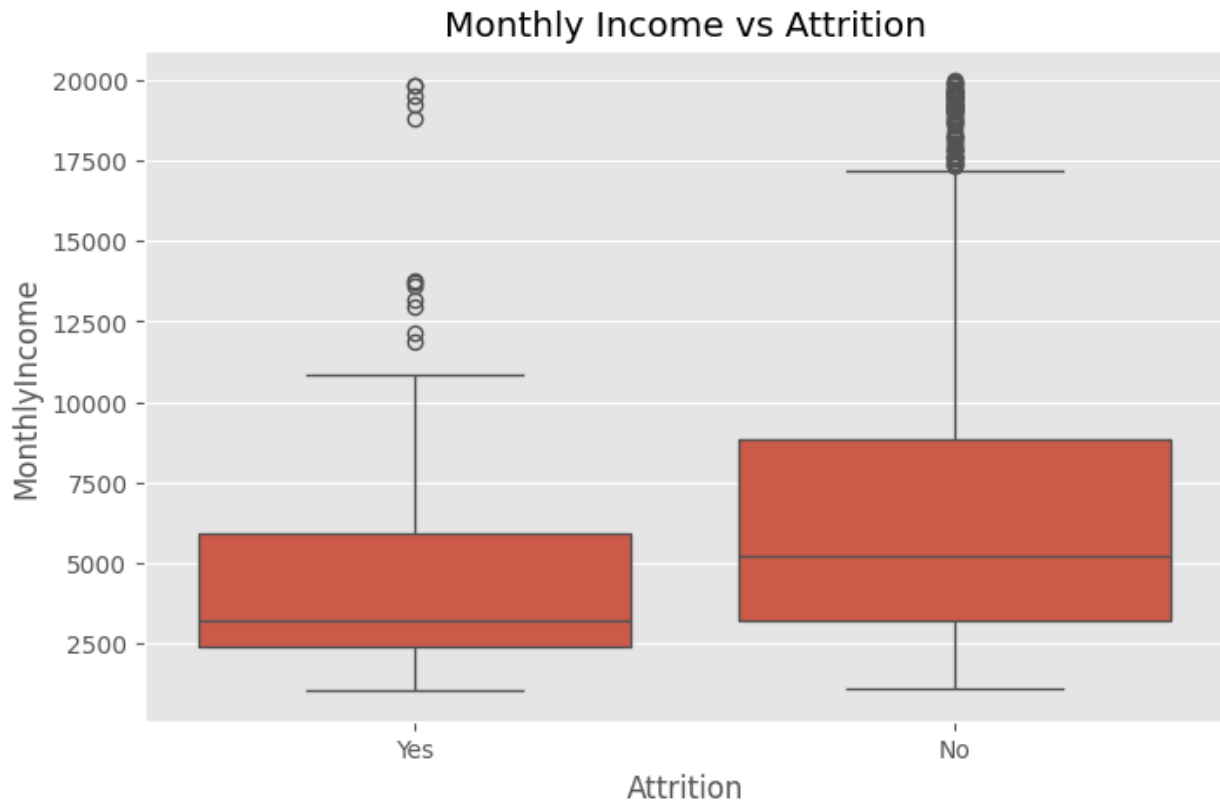
```
plt.title('Attrition by Job Role')
plt.show()
```



Sales Executives and Laboratory Technicians have the highest number of resignations, indicating these roles may require better retention strategies.

Does monthly income affect employee attrition?

```
plt.figure(figsize=(8,5))
sns.boxplot(data=df, x='Attrition', y='MonthlyIncome')
plt.title('Monthly Income vs Attrition')
plt.show()
```

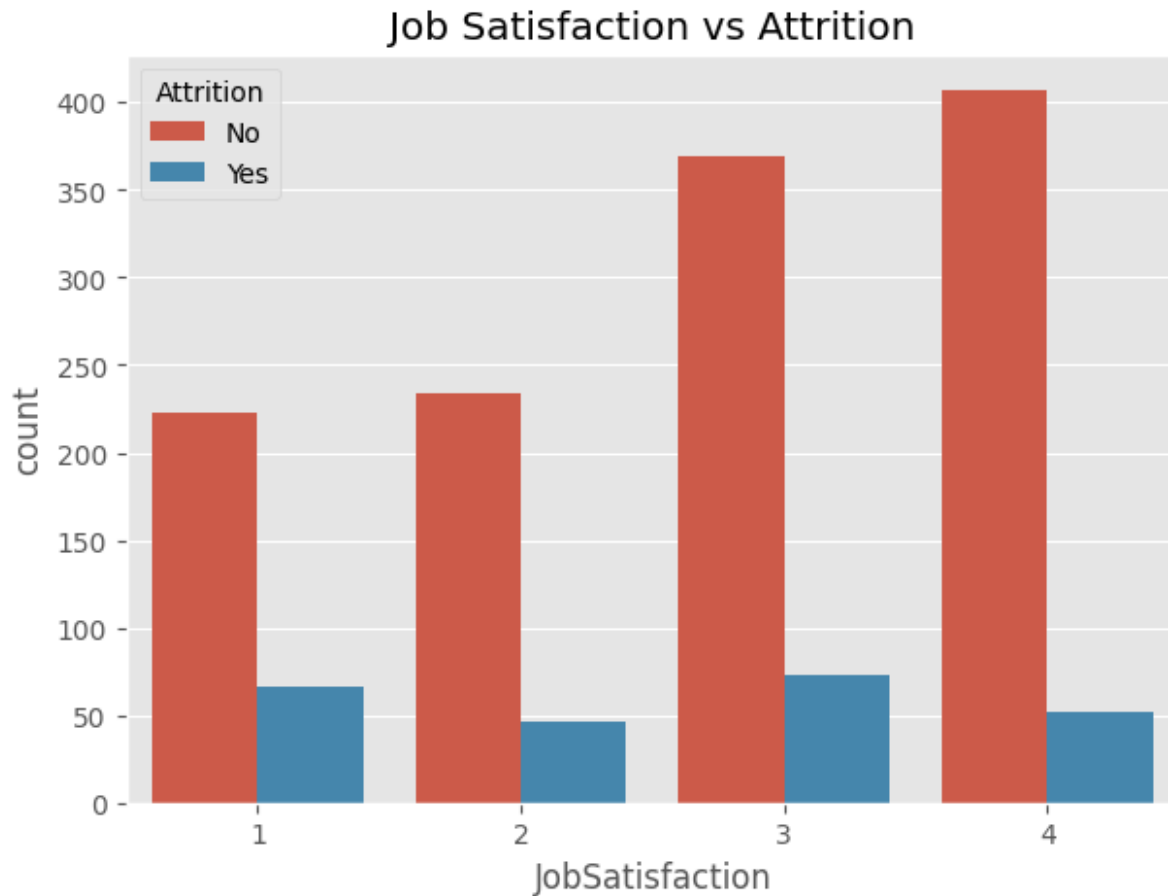


Employees who leave the organization generally have lower monthly incomes compared to employees who stay, suggesting that compensation may influence retention.

Does job satisfaction influence employee attrition?

```
plt.figure(figsize=(7,5))
sns.countplot(data=df,
              x='JobSatisfaction',
              hue='Attrition')

plt.title('Job Satisfaction vs Attrition')
plt.show()
```

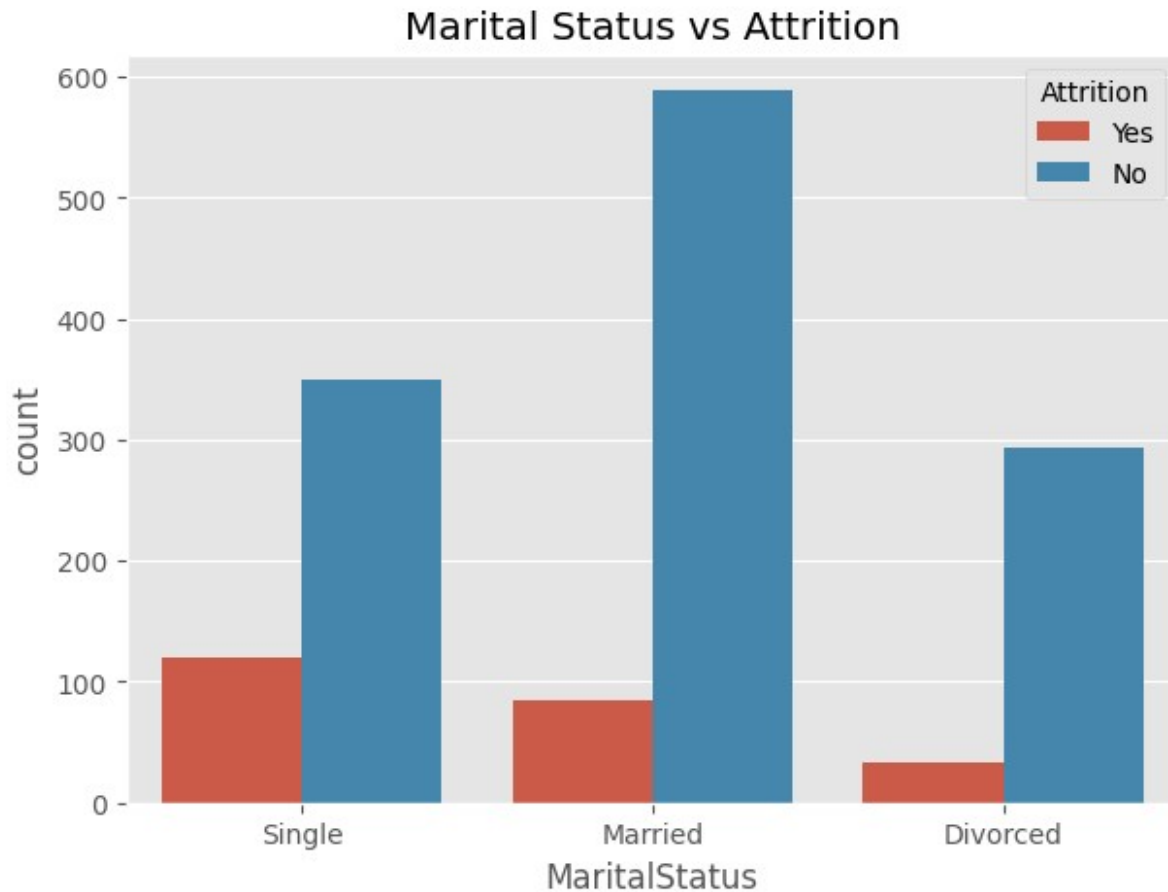


Employees with lower job satisfaction levels are more likely to leave the organization, highlighting the importance of maintaining employee engagement.

Which marital status has the highest employee attrition?

```
plt.figure(figsize=(7,5))
sns.countplot(data=df,
              x='MaritalStatus',
              hue='Attrition')

plt.title('Marital Status vs Attrition')
plt.show()
```

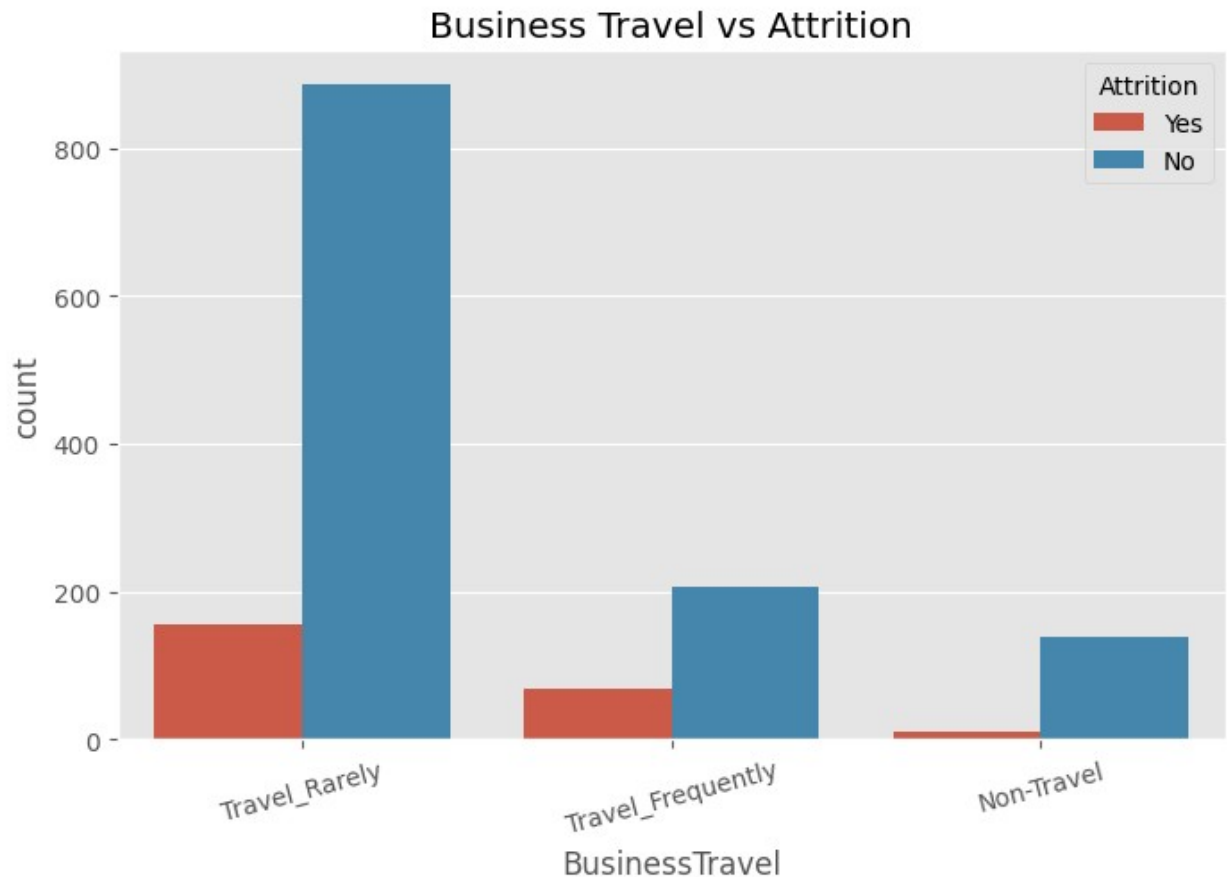


Single employees have the highest attrition, indicating they are more likely to switch jobs for career growth or personal reasons.

Does business travel affect employee attrition?

```
plt.figure(figsize=(8,5))
sns.countplot(data=df,
              x='BusinessTravel',
              hue='Attrition')

plt.title('Business Travel vs Attrition')
plt.xticks(rotation=15)
plt.show()
```



For a deeper analysis, we can calculate the attrition rate (%) within each travel category

```
# Calculate attrition rate by Business Travel

attrition_rate = (
    df.groupby('BusinessTravel')['Attrition']
      .apply(lambda x: (x == 'Yes').mean() * 100)
      .sort_values(ascending=False)
)

print(attrition_rate)

BusinessTravel
Travel_Frequently    24.909747
Travel_Rarely       14.956855
Non-Travel           8.000000
Name: Attrition, dtype: float64

plt.figure(figsize=(8,5))

sns.barplot(
```

```

    x=attrition_rate.index,
    y=attrition_rate.values,
    palette='viridis'
)

plt.title('Attrition Rate by Business Travel')
plt.xlabel('Business Travel')
plt.ylabel('Attrition Rate (%)')

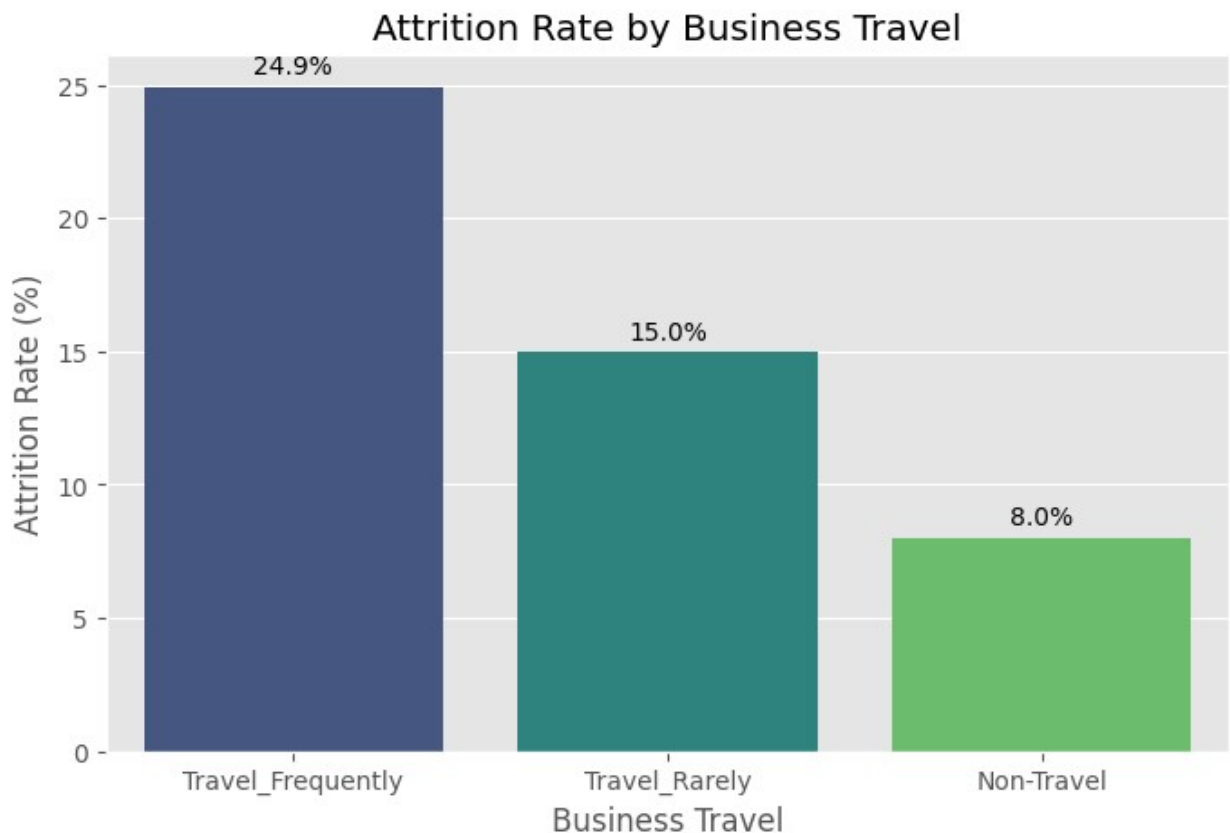
# Display percentage values on bars
for i, value in enumerate(attrition_rate.values):
    plt.text(i, value + 0.5, f'{value:.1f}%', ha='center')

plt.show()

/tmp/ipykernel_7919/225055738.py:3: FutureWarning:
Passing `palette` without assigning `hue` is deprecated and will be
removed in v0.14.0. Assign the `x` variable to `hue` and set
`legend=False` for the same effect.

sns.barplot(

```

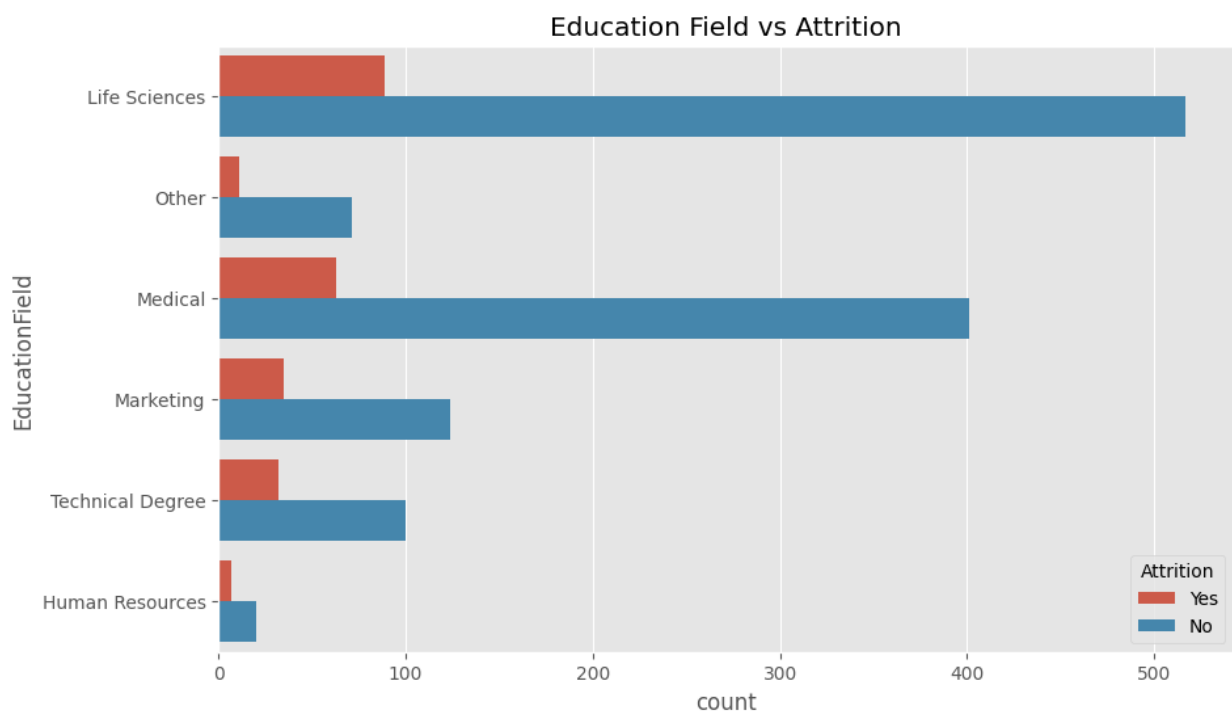


Although Travel_Rarely has the highest number of employees leaving the company, Travel_Frequently has the highest attrition rate (%). This indicates that employees who travel frequently are more likely to leave the organization, suggesting that extensive travel may negatively impact work-life balance and employee retention.

Which education field has the highest employee attrition?

```
plt.figure(figsize=(10,6))
sns.countplot(data=df,
              y='EducationField',
              hue='Attrition')

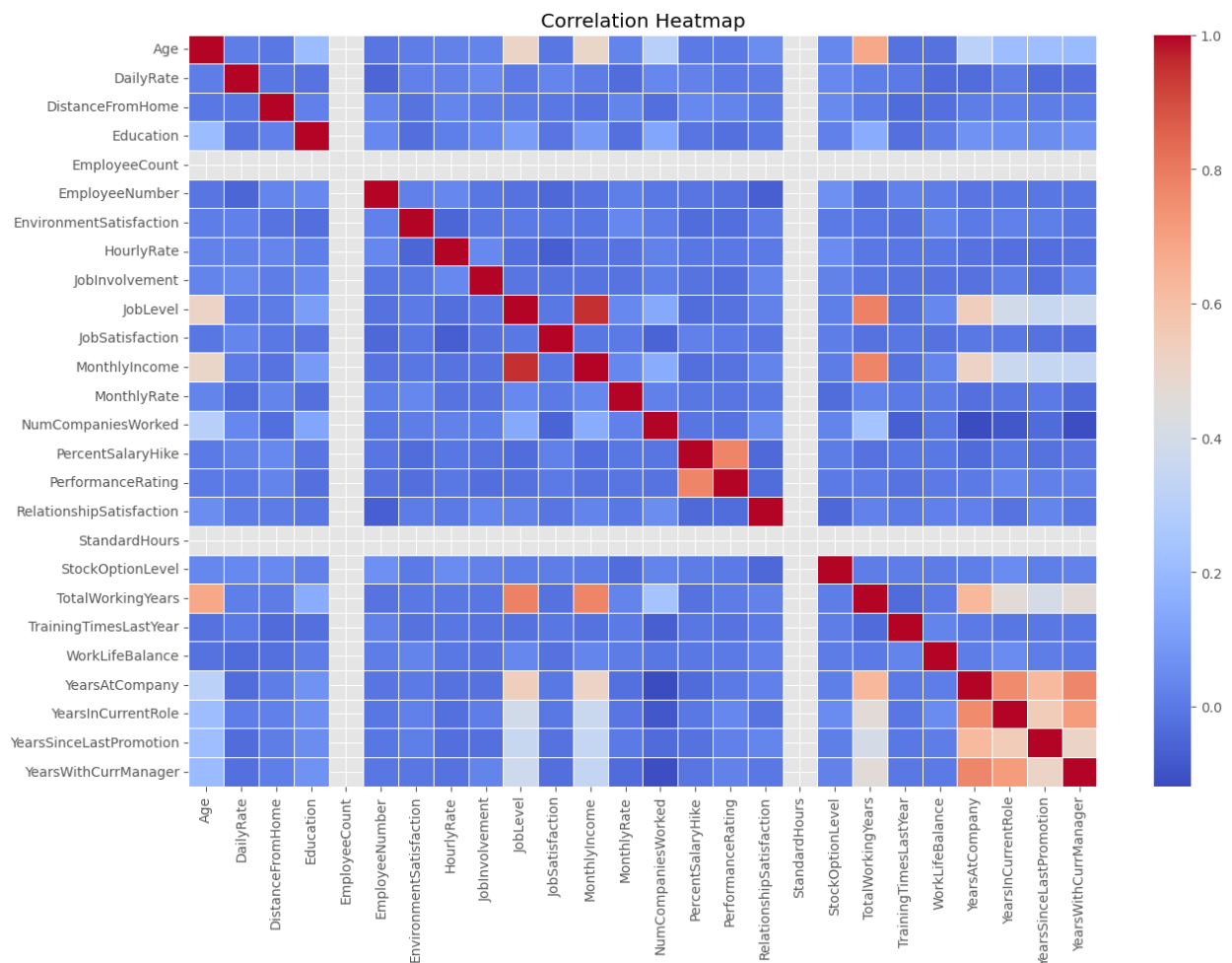
plt.title('Education Field vs Attrition')
plt.show()
```



Employees from the Life Sciences and Medical education field account for the highest number of attrition cases, reflecting both workforce size and turnover patterns.

Which factors are highly correlated?

```
plt.figure(figsize=(15,10))  
  
corr = df.corr(numeric_only=True)  
  
sns.heatmap(corr,  
            cmap='coolwarm',  
            linewidths=0.5)  
  
plt.title('Correlation Heatmap')  
plt.show()
```

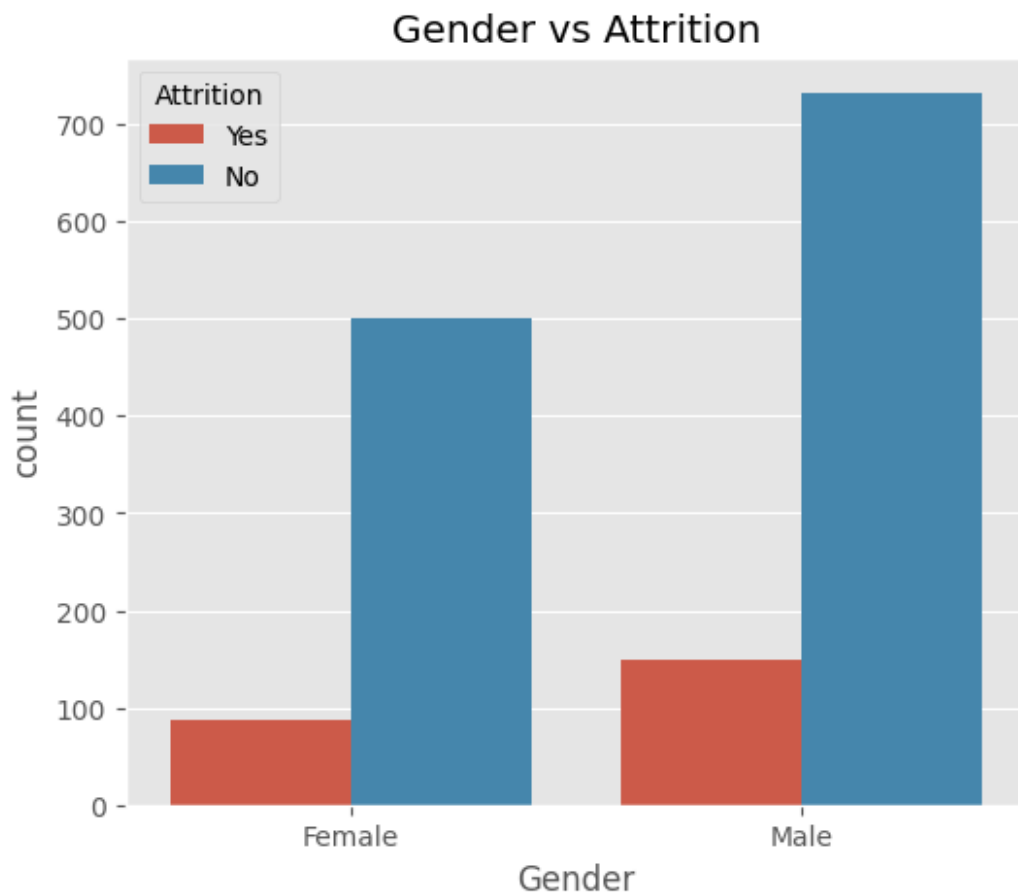


The heatmap shows relationships among numerical variables. Monthly Income has a positive relationship with Total Working Years and Job Level, while most variables have only weak linear relationships with attrition, suggesting turnover is influenced by multiple factors.

Does gender influence employee attrition?

```
plt.figure(figsize=(6,5))
sns.countplot(data=df,
              x='Gender',
              hue='Attrition')

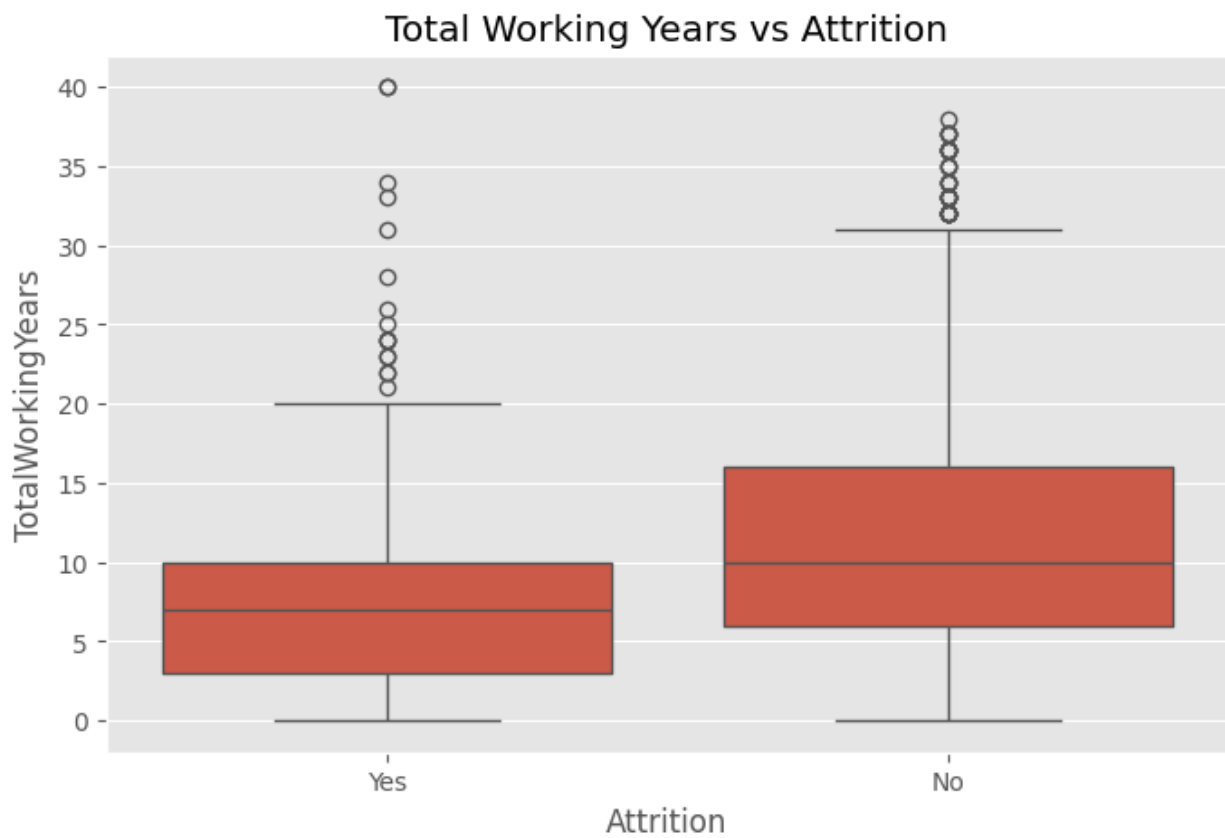
plt.title('Gender vs Attrition')
plt.show()
```



Both male and female employees experience attrition as compared to their total numbers

```
plt.figure(figsize=(8,5))
sns.boxplot(data=df,
            x='Attrition',
            y='TotalWorkingYears')

plt.title('Total Working Years vs Attrition')
plt.show()
```



Employees with fewer years of work experience are generally more likely to leave the organization, indicating that early-career employees may require additional support and development opportunities.

Final Business Recommendations

Based on the analysis, the organization should consider the following recommendations:

- 1-Develop targeted retention strategies for the Research & Development (R&D) department, as it has the highest number of attrition cases.
- 2-Review workload and overtime policies to improve work-life balance.
- 3-Strengthen employee engagement and job satisfaction through regular feedback, recognition, and career development initiatives.
- 4-Review compensation structures, particularly for lower-income employees, to remain competitive.
- 5-Design mentoring and career progression programs for younger employees and those with fewer years of experience.
- 6-Monitor high-turnover job roles such as Laboratory Technicians and Sales Executives and investigate their specific challenges.
- 7-Evaluate the impact of business travel on employee well-being and introduce flexible work arrangements where possible.
- 8-Build an HR analytics dashboard to continuously monitor attrition trends and support data-driven decision-making.

Conclusion

This Exploratory Data Analysis identified key factors associated with employee attrition, including department, age, overtime, job role, monthly income, job satisfaction, marital status, business travel, education field, gender, and work experience. The findings indicate that attrition is influenced by a combination of workplace conditions and employee characteristics rather than a single factor. By implementing targeted retention strategies and continuously monitoring workforce metrics, the organization can improve employee satisfaction, reduce turnover, and build a more stable and productive workforce.